

Name: _____

Answer all questions below.

1. Assume I want each thread to execute the following critical section, where counter is a variable shared between all threads.

```
for (int i = 0; i < 100; i++)
{
    if (...)
        counter = counter + 1;
}
```

- (a) What are the pros and cons of placing a lock outside of the for loop?
 - (b) What are the pros and cons of placing a lock within the for loop?
 - (c) How would you rewrite this code to optimize the pros and cons listed above?
2. Assume I have a large number of threads. How would you change the semaphore below to guarantee it will work with many threads?

```
wait(sem_t* sem)
{
    while (sem->S <= 0) {} // busy wait
    sem->S = sem->S - 1;
}

signal(sem_t* sem)
{
    sem->S++;
}
```